

 CISCO NETWORKING ACADEMY & IILM UNIVERSITY

Introduction to Modern AI

Internship & Training Report on Core Concepts, Deep Learning, Generative AI, and
Ethical Governance

PRESENTED BY
Mubarak Alik

ROLL NUMBER
CS-23411290

DEGREE & BATCH
B.Tech CSE (2023–2027)

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Program Profile & Official Credential

Credential Field	Verified Information
Learner Name	Mubarak Alik (Roll No: CS-23411290)
Course Name	Introduction to Modern AI
Offering Institution	IILM University, School of Computer Science & Engineering
Academy Partner	Cisco Networking Academy Program
Course Instructor	Swati Vashisht
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| Training Purpose & Core Objectives



AI Foundations

Develop a clear conceptual understanding of Artificial Intelligence terminology, modern architectures, and core paradigms.



Data & Learning

Understand how supervised, unsupervised, and reinforcement algorithms extract patterns from structured and unstructured data.



Deep Learning & LLMs

Examine multi-layer neural networks, generative AI, transformer models, and prompt-driven decision tools.



Responsible AI

Identify critical ethical considerations including algorithmic bias, privacy, security risks, transparency, and human oversight.

| Core Subfields of Modern AI



Machine Learning & Vision

Machine Learning: Focuses on statistical algorithms that learn patterns from data to make predictions without direct rule programming.

Computer Vision: Enables computers to derive meaningful information from visual inputs such as digital images and videos.



NLP & Generative AI

Natural Language Processing: Allows machines to analyze, understand, interpret, and manipulate human text and speech seamlessly.

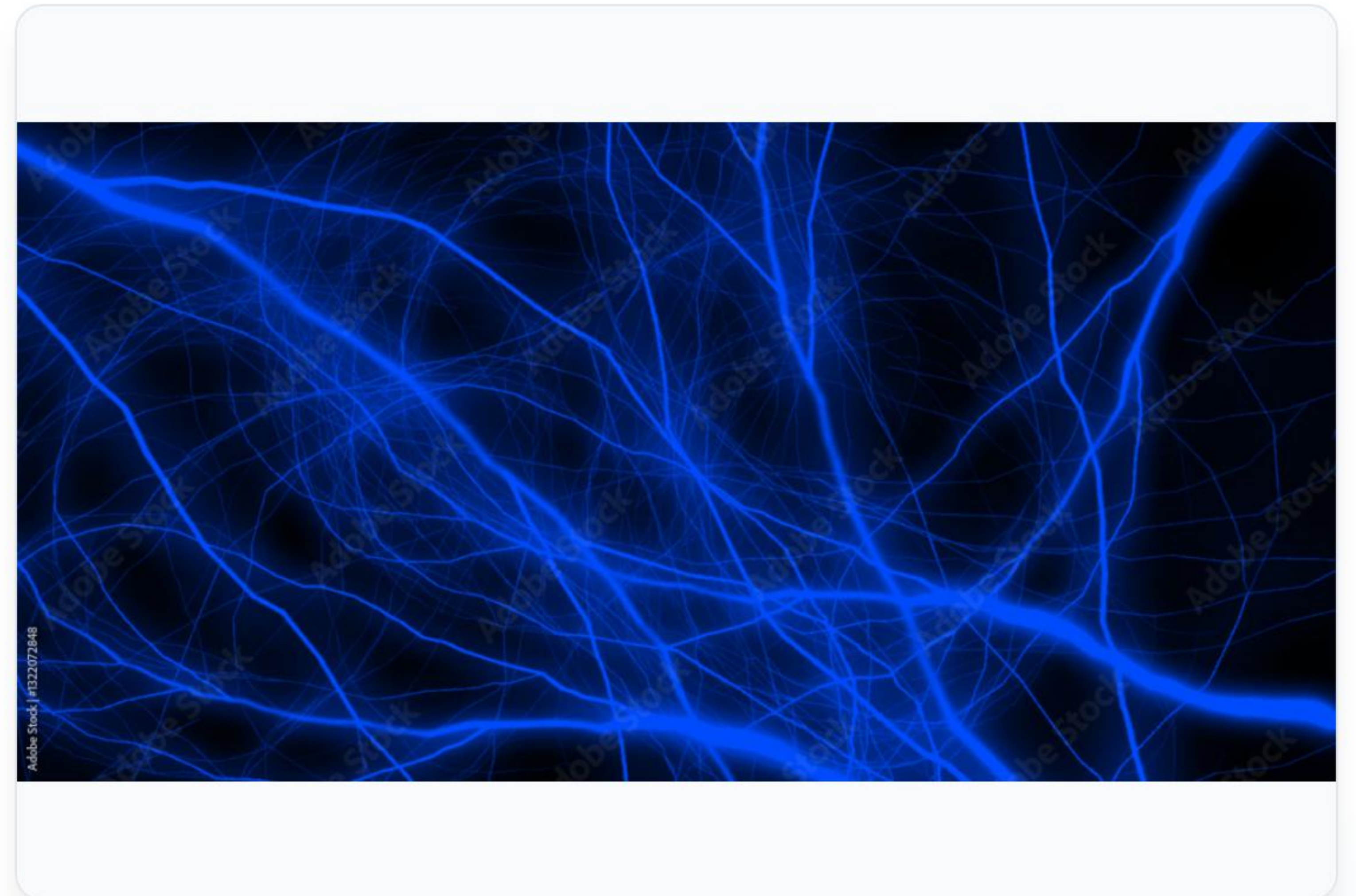
Generative AI: Utilizes deep generative models to produce novel text, synthetic images, code, and audio based on user prompts.

| Taxonomy of Machine Learning

Learning Type	Mechanism & Signal	Primary Goal	Practical Example
Supervised Learning	Learns from labeled inputs and corresponding target outputs.	Map input variables to target class or value.	Email spam detection, credit scoring, medical diagnosis.
Unsupervised Learning	Processes unlabeled data to uncover hidden structure.	Discover patterns, clusters, and dimensionality.	Customer segmentation, anomaly detection, topic modeling.
Reinforcement Learning	Agent interacts with environment via action, state, reward.	Maximize cumulative reward over time.	Autonomous robotics, game agents (AlphaGo), navigation.

Machine Learning Workflow & Data

- ✓ **Problem Definition:** Establish clear measurable targets and domain metrics.
- ✓ **Data Pipeline:** Collect, clean, normalise, and engineer relevant features.
- ✓ **Training & Validation:** Split dataset to fit model and tune hyperparameters.
- ✓ **Overfitting Management:** Prevent models from memorizing noise via regularization and cross-validation.

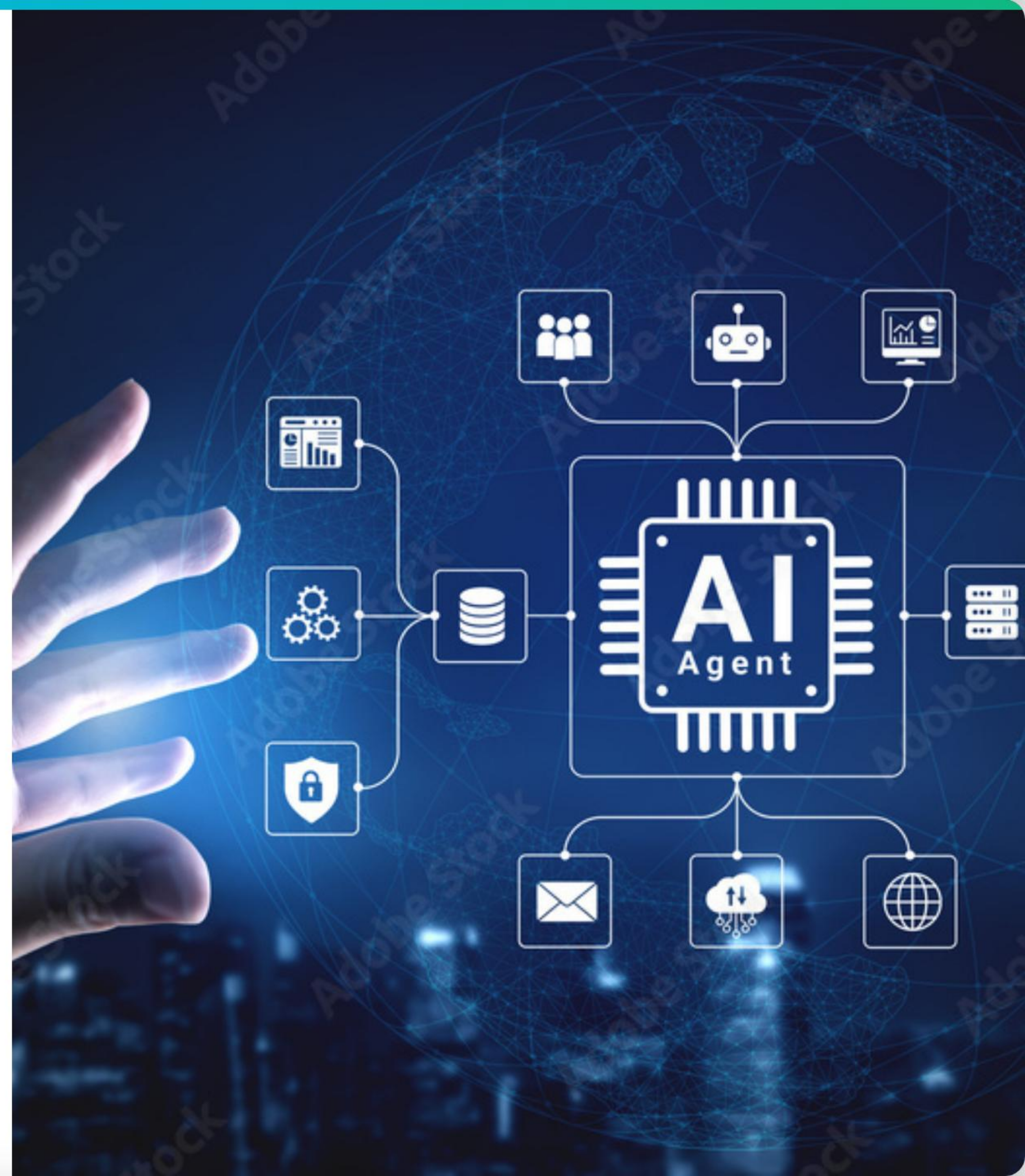


Deep Learning & Generative AI

Neural Networks at Scale

Deep learning leverages interconnected artificial neural networks with multiple layers to extract hierarchical representations from massive datasets.

Generative AI and Large Language Models (LLMs) build upon these deep architectures to process context, understand syntax, and generate accurate synthetic content across code, text, and visual formats.



LLMs & Human-in-the-Loop AI



Statistical Language Models

LLMs learn token distribution probabilities across vast text corpora, enabling automated drafting, summarization, and translation.



Verification & Hallucinations

Fluent language generation does not guarantee factual correctness. Outputs require continuous domain verification and sanity testing.



Human-AI Collaboration

Treating AI as an intelligent assistant where human experts define objectives, review outputs, and retain final accountability.

Real-World AI Applications

Domain	Practical AI Integration & Impact
Healthcare	Medical image classification, diagnostic decision support, patient intake triage.
Finance	Real-time transaction fraud detection, credit risk assessment, automated trading signals.
Cybersecurity	Automated anomaly detection, network threat hunting, security log analysis.
Retail & Supply	Personalized recommendation engines, inventory forecasting, route optimization.
Software Engineering	Code completion assistance, automated unit test generation, documentation tooling.

Pillars of Responsible AI

-  **Fairness & Mitigation of Bias:** Auditing training datasets to prevent discriminatory outcomes and unequal predictions.
-  **Privacy & Data Protection:** Safeguarding personally identifiable information (PII) through strict compliance and data governance.
-  **Transparency & Explainability:** Ensuring model decisions are explainable to end-users and stakeholders.
-  **Security & Robustness:** Protecting models against adversarial prompt injections and data poisoning attacks.
-  **Accountability & Oversight:** Maintaining human control over critical decisions in high-stakes environments.

| Learning Outcomes & Growth

100%

Course Completion

8

Core Modules
Mastered

Key Competencies Developed

- ✓ **Analytical Thinking:** Ability to decompose AI solutions into data, model, and validation stages.
- ✓ **Technical Literacy:** Clear comprehension of ML, Neural Networks, Transformer architectures, and generative prompts.
- ✓ **Governance & Ethics:** Practical understanding of NIST AI Risk Management Framework principles.

Thank You!

Introduction to Modern AI — Training & Internship Presentation



Student: Mubarak Alik



Roll Number: CS-23411290



Institution: IILM University, Greater Noida



Program: Cisco Networking Academy



Future Roadmap: Hands-on ML with Python & PyTorch

| Image Sources



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